

1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

After opening a typical desktop cabinet, the following major components may be found. Exact hardware varies by computer model and some parts, such as a dedicated graphics card or optical drive, may be optional.

1. Main Components Inside a Desktop PC



No.	Component	Short description
1	Power Supply Unit (PSU)	Converts AC power from the wall outlet into regulated DC power used by the motherboard, drives and other components.
2	Motherboard	The main circuit board. It connects the CPU, RAM, storage, expansion cards and other devices and allows them to communicate.
3	CPU (Processor)	Performs instructions and calculations for the operating system and applications. It is normally installed in a socket on the motherboard.
4	CPU Cooler	Removes heat from the CPU using a heatsink and fan or another cooling system, helping the processor operate safely.
5	RAM (Memory)	Temporary working memory used by the CPU for currently running programs and data. It is installed in motherboard memory slots.
6	HDD / SSD	Permanent storage for the operating system, applications and user files. HDDs use magnetic disks; SSDs use flash memory.
7	Graphics Card (GPU)	An expansion card that processes graphics and video. Some computers instead use graphics integrated into the CPU or motherboard platform.
8	Cables / Connectors	Carry power and data between components, such as motherboard power, SATA data, front-panel and GPU power cables.
9	Optical Drive (optional)	Reads or writes CDs/DVDs. It is less common in modern desktop PCs.
10	Computer Case / Chassis	The enclosure that supports, protects and organizes the internal components while providing airflow and access points.

2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step. Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

2. Disassembly Steps (Step-by-Step)

1 Unplug Power  Turn off the PC and unplug the power cable from the wall and PSU. This prevents electric shock and protects components.	2 Remove Side Panel  Unscrew the side panel (usually on the left) and remove it to access the internal components.	3 Disconnect Cables  Carefully unplug power cables, SATA cables, front panel connectors and other wiring from the motherboard, GPU and drives.	4 Remove RAM  Push the side clips outward and gently remove the RAM modules.	5 Remove Storage (HDD/SSD)  Disconnect SATA and power cables, then unscrew and remove the HDD/SSD from its bay.	6 Remove Graphics Card (GPU)  Unscrew the bracket and gently pull the GPU out of the PCIe slot.	7 Remove Motherboard  Unscrew the motherboard standoffs and remove the board from the case.
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Step 1 — Shut down and unplug power

What I did: Turn off the computer normally, switch the PSU off if it has a switch, unplug the AC power cable, and disconnect external peripherals. Press the power button briefly after unplugging to help discharge residual power.

Why: This prevents electric shock and reduces the chance of short circuits or component damage.

Step 2 — Remove the side panel

What I did: Place the cabinet on a stable surface. Remove the rear panel screws or thumbscrews and slide/lift the side panel away as designed by the case.

Why: This provides safe access to the internal components.

Step 3 — Disconnect internal cables

What I did: Before pulling anything, identify and disconnect the 24-pin motherboard power connector, CPU power connector, SATA/data cables, GPU power cables and front-panel connectors as applicable.

Why: Disconnecting cables first prevents cables from snagging when components are removed. Do not pull on wires; grip the connector.

Step 4 — Remove RAM modules

What I did: Open the retaining clips at the ends of the RAM slot. Hold the module by its edges and lift it straight out. Place it in an anti-static bag or on a safe non-conductive surface.

Why: RAM is sensitive to electrostatic discharge, and the clips/contacts can be damaged by rough handling.

Step 5 — Remove HDD/SSD

What I did: Disconnect SATA power and data cables if present. Remove the drive from its tray/caddy or mounting screws. M.2 SSDs are removed by taking out the retaining screw and lifting the module at an angle.

Why: Removing storage makes the remaining motherboard/cable work easier and prevents the drive from being damaged during motherboard removal.

Step 6 — Remove the graphics card (if installed)

What I did: Remove any PCIe power connector. Unscrew the card's rear bracket, release the PCIe slot latch, and gently pull the card straight out while supporting it.

Why: The slot latch and connector must be released before lifting the card to avoid physical damage.

Step 7 — Remove the motherboard

What I did: Disconnect any remaining cables. Remove the motherboard mounting screws, support the board with both hands, and lift it clear of the case while avoiding contact with exposed circuitry.

Why: The motherboard should be removed only after attached cables and expansion cards have been cleared.

3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

3. Reassembly Steps (Step-by-Step)

1 Install Motherboard



Place the motherboard in the case and secure it with standoffs and screws.

2 Install CPU & Cooler



Ensure the CPU is seated correctly and mount the cooler (with thermal paste if required).

3 Install RAM



Insert RAM modules into the slots until the clips lock.

4 Install Storage (HDD/SSD)



Mount the drive in its bay and connect SATA data and power cables.

5 Install Graphics Card (GPU)



Insert the GPU into the PCIe slot and secure it with screws.

6 Connect Cables



Reconnect all power, data and front panel cables (USB, audio, etc.).

7 Close Case & Power On



Attach the side panel, plug in the power cable and turn on the PC. Check that everything works properly.

Step 1 — Install the motherboard

What I did: Position the motherboard over the correct case standoffs and align the rear I/O area. Secure it with the correct screws without overtightening.

Why the order matters: The motherboard forms the platform for the CPU, RAM, expansion cards and most connections, so it should be secured before other parts that may obstruct access.

Step 2 — Install CPU and cooler

What I did: If removed, install the CPU in the socket according to its alignment mark, apply the manufacturer's recommended thermal interface material when required, and secure the CPU cooler.

Why the order matters: The CPU must be correctly seated and cooled before the system is powered on.

Step 3 — Install RAM

What I did: Align the notch in each RAM module with the slot key and press evenly until the retaining clips lock into position.

Why the order matters: Correct alignment prevents damage to the module and slot. RAM should be fully seated before testing the system.

Step 4 — Install HDD/SSD

What I did: Mount the storage drive in its tray/caddy or M.2 position. Secure it and connect the required power and data cables.

Why the order matters: Storage must be physically secure and connected so the system can access the operating system and files.

Step 5 — Install the graphics card

What I did: Insert the GPU into the appropriate PCIe slot until the latch engages, secure its bracket to the case and connect any required GPU power cable.

Why the order matters: The card must be fully seated and mechanically secured for reliable operation.

Step 6 — Reconnect all cables

What I did: Connect the 24-pin motherboard power, CPU power, SATA/data, GPU power and front-panel connectors as applicable. Route cables so they do not obstruct fans.

Why the order matters: Correct power and data connections are essential for startup and stable operation. Cable routing also helps airflow.

Step 7 — Close the case and test

What I did: Check that no tools or loose screws remain inside. Replace the side panel, reconnect the external cables, switch on the PSU if applicable, and start the PC.

Why the order matters: Testing after a final inspection reduces the risk of powering the system with a loose connector, foreign object or obstructed fan.

4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

A. Safety Precautions

Precaution	Why it is important
Turn off and unplug the PC	Prevents electrical shock and reduces the risk of short circuits while components are being handled.
Use a clean, dry, well-lit work area	Reduces the chance of losing small parts and makes connectors, screws and labels easier to see.
Prevent electrostatic discharge (ESD)	Use an anti-static wrist strap if available and touch/handle components appropriately. ESD can damage sensitive electronics even when no visible damage occurs.
Handle components by their edges	Avoids touching contacts, pins and exposed circuitry and reduces the chance of bending or contaminating contacts.
Do not force connectors or parts	Most connectors have a key, latch or orientation. Forcing them can break clips, pins or sockets.
Keep screws organized	Different screws may have different lengths. Mixing them can cause incorrect assembly or damage to the case/component.
Keep liquids and food away	Prevents spills and contamination near electrical equipment.
Never work inside a powered-on PC	Fans, electrical circuits and exposed connections can create hazards and accidental component damage.
Allow hot components to cool	A recently running PC may contain hot heatsinks or other parts that can cause burns.
Follow the motherboard/case manual	Connector locations, mounting points and installation procedures vary between models.

B. Tools and Materials

Tool / material	Purpose
Phillips screwdriver	Used for most motherboard, drive and case screws.
Flat-head screwdriver (optional)	May be useful for compatible screws or carefully releasing certain clips; use only where appropriate.
Anti-static wrist strap	Helps reduce electrostatic discharge risk when working with sensitive components.
Small container / screw tray	Keeps screws and small parts organized during disassembly.
Cable ties / Velcro straps	Helps route and secure cables during reassembly.
Thermal paste (if required)	Used when reinstalling a CPU cooler if the existing thermal interface must be replaced.
Soft, clean work surface	Provides a stable area for components and reduces scratching or accidental contact.